Identification of Model 5

This document analyzes the identification of the structural logit model with information asymmetry and switching costs developed in `model5_convert4intostructural_v3.tex`.

1 Model Recap

Two banks compete for a borrower. Bank 1 (informed/incumbent) observes the borrower's default probability h; Bank 2 (uninformed/entrant) does not. Utility:

$$u_{i1} = -\alpha\, r_{i1} + \varepsilon_{i1}, \qquad u_{i2} = -\alpha\, r_{i2} - \lambda + \varepsilon_{i2}$$

with ε_{ij} i.i.d. Type I extreme value, giving logit market shares:

$$s(r_1, r_2) = \frac{1}{1 + \exp\big(\alpha(r_1 - r_2) - \lambda\big)} \tag{1}$$

In equilibrium, bank 1 plays a pure strategy $\sigma^*(h)$ and bank 2 plays a single rate r_2^*, characterized by:

$$\sigma^*(h) = \frac{1}{1-h} + \frac{1}{\alpha\big(1 - s(\sigma^*(h), r_2^*)\big)} \tag{2}$$

$$r_2^* = \frac{1}{1-\tilde{h}} + \frac{1}{\alpha} \cdot \frac{\int (1-s)(1-h)\, dF}{\int s\,(1-s)(1-h)\, dF} \tag{3}$$

Key equilibrium properties: $\sigma^*(h)$ is strictly increasing and the markup $\sigma^*(h) - \frac{1}{1-h}$ is decreasing in h.

2 Estimation Primitives

The structural parameters to be estimated are:

Parameter	**Description**	**Dimension**
α	Price sensitivity	Scalar
λ	Switching cost	Scalar
$F(\cdot)$	CDF of default probabilities on $[h_{\min}, h_{\max}]$	Function

In a parametric approach, F is specified by a finite-dimensional parameter (e.g., Kumaraswamy(a, b) on $[h_{\min}, h_{\max}]$), so the full parameter vector is $\theta = (\alpha, \lambda, a, b, h_{\min}, h_{\max})$. We first establish non-parametric identification—where F is treated as an infinite-dimensional object—and then discuss parametric estimation.

All other model objects are <u>derived</u> from these primitives:

- The equilibrium strategy $\sigma^*(h)$ is determined by (α, λ, F) through the system (2)–(3).
- Market shares $s(\sigma^*(h), r_2^*)$, markups, default rates, and profits follow.

3 Observable Data

For each borrower $i = 1, \ldots, N$, we observe:

Variable	Description	Values
r_i^{win}	Interest rate charged by the chosen bank	$\mathbb{R}_+$
W_i	Identity of the chosen bank	$\{1, 2\}$
D_i	Default indicator	$\{0, 1\}$

What is NOT observed:

1. The borrower's true default probability h_i (latent type)
2. The losing bank's rate offer
3. Bank profits

Remark 1 (Data structure). When bank 1 wins ($W_i = 1$), the winning rate $r_i^{\text{win}} = \sigma^*(h_i)$ varies across borrowers because bank 1 conditions on h_i. When bank 2 wins ($W_i = 2$), the winning rate $r_i^{\text{win}} = r_2^*$ is the <u>same</u> for all bank 2 winners. This asymmetry is a direct consequence of the information structure and is a distinctive feature of the data.

4 Identification Argument

We establish identification sequentially: each step uses objects identified in previous steps.

Theorem 2 (Identification)**.** *The structural parameters (α, λ, F) are identified from the joint distribution of (r_i^{win}, W_i, D_i).*

The proof proceeds in four steps.

4.1 Step 1: Identification of r_2^*

Proposition 3. *The entrant's rate r_2^* is identified from the data.*

Proof. Bank 2 plays a single rate for all borrowers. When $W_i = 2$, the winning rate is $r_i^{\text{win}} = r_2^*$. In the data, r_2^* is therefore the common value of r_i^{win} among all observations with $W_i = 2$. □

4.2 Step 2: Identification of $\sigma(\cdot)$ and the Mapping $h \leftrightarrow r_1$

Proposition 4. *The informed bank's pricing function $\sigma(\cdot)$ and its inverse $\tau(\cdot) = \sigma^{-1}(\cdot)$ are identified from (r_i^{win}, D_i) conditional on $W_i = 1$.*

Proof. When bank 1 wins, the observed rate is $r_i^{\text{win}} = \sigma^*(h_i)$ and the default outcome is $D_i \sim \text{Bernoulli}(h_i)$. Since default is conditionally independent of other variables given h_i:

$$E[D_i \mid r_i^{\text{win}} = r,\ W_i = 1] = h_i = \tau(r) \tag{4}$$

This identifies the inverse pricing function: the conditional default rate at each observed rate r among bank 1 winners equals the default probability of the type that receives rate r. Since σ^* is strictly monotone (by the equilibrium characterization), $\tau = \sigma^{-1}$ is well-defined.[1] □

Remark 5 (No functional form needed). Step 2 does not require any parametric assumption on F, σ, or the equilibrium. It uses only the fact that default probability equals the conditional expectation of default, and that σ is invertible.

After Step 2, for each bank 1 winner we can compute:

- $h_i = \tau(r_i^{\text{win}})$: the inferred type
- $m_i \equiv r_i^{\text{win}} - \frac{1}{1-h_i} = \sigma^*(h_i) - c(h_i)$: the observed <u>markup</u>
- $\Delta_i \equiv r_i^{\text{win}} - r_2^* = \sigma^*(h_i) - r_2^*$: the <u>rate gap</u>

These are data objects, computed without knowledge of α or λ.

4.3 Step 3: Joint Identification of α and λ

This is the key step. The equilibrium FOC imposes a restriction linking the data objects $m(h)$ and $\Delta(h)$ to the structural parameters α and λ.

Proposition 6. *The parameters α and λ are jointly identified from the pricing function $\sigma^*(\cdot)$, the break-even rate $c(h) = \frac{1}{1-h}$, and the entrant rate r_2^*.*

Proof. **Deriving the identifying restriction.** From the FOC (2), the markup satisfies:

$$m(h) = \sigma^*(h) - \frac{1}{1-h} = \frac{1}{\alpha(1-s)}$$

Substituting the logit share $1 - s = \frac{e^{\alpha\Delta(h)-\lambda}}{1+e^{\alpha\Delta(h)-\lambda}}$:

$$m(h) = \frac{1 + e^{\alpha\Delta(h)-\lambda}}{\alpha\, e^{\alpha\Delta(h)-\lambda}} = \frac{1}{\alpha}\left(e^{\lambda-\alpha\Delta(h)} + 1\right)$$

Rearranging:

$$\alpha\, m(h) - 1 = e^{\lambda - \alpha\,\Delta(h)} \tag{5}$$

Taking logarithms:

$$\boxed{\ln\big(\alpha\, m(h) - 1\big) = \lambda - \alpha\,\Delta(h)} \tag{6}$$

[1] In finite samples, $E[D|r, W = 1]$ is estimated nonparametrically, e.g., by kernel regression of D_i on r_i^{win} among bank 1 winners.

This must hold for every type $h \in [h_{\min}, h_{\max}]$. Both $m(h)$ and $\Delta(h)$ are known data objects (from Steps 1–2), while α and λ are the two unknowns.

Separation of α and λ. Evaluating (6) at any two distinct types $h_a \neq h_b$ and subtracting:

$$\ln \frac{\alpha\, m(h_a) - 1}{\alpha\, m(h_b) - 1} = \alpha \left(\Delta(h_b) - \Delta(h_a)\right) \tag{7}$$

This is a single equation in the single unknown α, since $m(h_a), m(h_b), \Delta(h_a), \Delta(h_b)$ are all data.

Uniqueness. Define $\psi(\alpha) \equiv \ln(\alpha\, m_a - 1) - \ln(\alpha\, m_b - 1) - \alpha(\Delta_b - \Delta_a)$ where we abbreviate $m_j = m(h_j)$, $\Delta_j = \Delta(h_j)$. Without loss of generality take $h_a < h_b$, so that $m_a > m_b$ (markups are declining) and $\Delta_a < \Delta_b$ (σ^* is increasing). We verify:

- Domain: We need $\alpha m_j > 1$ for both j. From the equilibrium, $\alpha\, m(h) = \frac{1}{1-s(h)} > 1$ always, so this holds for the true α and indeed for any $\alpha > 1/m_b$ (the more binding constraint since $m_b < m_a$).
- Boundary behavior: As $\alpha \downarrow 1/m_b$: $\ln(\alpha m_b - 1) \to -\infty$ while $\ln(\alpha m_a - 1)$ remains finite, so $\psi(\alpha) \to +\infty$. As $\alpha \to \infty$: $\ln(\alpha m_a - 1) - \ln(\alpha m_b - 1) \to \ln(m_a/m_b)$, a finite constant, while $\alpha(\Delta_b - \Delta_a) \to +\infty$, so $\psi(\alpha) \to -\infty$.
- Continuity: ψ is continuous on $(1/m_b, \infty)$.

By the intermediate value theorem, ψ has at least one root. Moreover, ψ is strictly decreasing: $\psi'(\alpha) = \frac{m_a}{\alpha m_a - 1} - \frac{m_b}{\alpha m_b - 1} - (\Delta_b - \Delta_a)$. The function $m \mapsto \frac{m}{\alpha m - 1}$ is strictly decreasing (its derivative is $\frac{-1}{(\alpha m - 1)^2} < 0$), so $m_a > m_b$ implies $\frac{m_a}{\alpha m_a - 1} < \frac{m_b}{\alpha m_b - 1}$, making the first difference negative; subtracting the positive term $\Delta_b - \Delta_a$ gives $\psi' < 0$.

Therefore the root is unique, identifying α.

Recovery of λ. Given α, substitute into (6) at any h:

$$\lambda = \ln\big(\alpha\, m(h) - 1\big) + \alpha\, \Delta(h) \tag{8}$$

The fact that this must be constant across all h provides overidentifying restrictions (Section 5). □

Remark 7 (Intuition for separate identification). Why can α and λ be told apart?

- α **governs the slope.** Higher price sensitivity means bank 1's markup responds more steeply to the rate gap $\Delta(h)$. In (6), α is the slope of $\ln(\alpha m - 1)$ as a function of Δ.
- λ **governs the level.** Switching costs shift the intercept: higher λ raises bank 1's markup for all types uniformly (by reducing the effective competitiveness of bank 2). In (6), λ is the intercept.

Identification exploits the fact that $\sigma^*(h)$ varies across types (generating variation in m and Δ), while λ does not. Two types suffice to pin down the slope (α); the level (λ) then follows.

Remark 8 (Comparison to Model 4 identification). In model 4 (without logit demand), identification of λ relied on the clean relationship $\lambda = r_1^{\min} - r_2^{\min}$: the gap between the lowest incumbent and entrant rates. Here the identification logic is different: there is no "lowest entrant rate" (bank 2 plays a single rate), and λ is not simply a rate gap. Instead, λ is identified from the curvature of the markup schedule — how the informed bank's margin varies with the rate gap — which requires at least two type observations.

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4.4 Step 4: Identification of $F(h)$

Proposition 9. *The type distribution $F(h)$ is nonparametrically identified.*

Proof. With α, λ known (from Step 3), the market share for each type is:

$$s(h) \equiv s(\sigma^*(h), r_2^*) = \frac{1}{1 + \exp\big(\alpha\,\Delta(h) - \lambda\big)} \tag{9}$$

which is a known function of h (since σ^* and r_2^* are identified).

Among bank 1 winners, the density of types is the selection-weighted density:

$$f_{h|W=1}(h) = \frac{s(h)\,f(h)}{\Pr(W_i = 1)} = \frac{s(h)\,f(h)}{\int s\,dF} \tag{10}$$

The left side is estimable from the data: among bank 1 winners, we recover $h_i = \tau(r_i^{\text{win}})$ (Step 2), and the empirical distribution of these h_i values estimates $f_{h|W=1}$.

Inverting (10):

$$f(h) = \frac{f_{h|W=1}(h) \cdot \Pr(W_i = 1)}{s(h)} \tag{11}$$

Since $s(h) > 0$ for all h (logit shares are strictly interior), this is well-defined. The aggregate probability $\Pr(W_i = 1) = \frac{N_1}{N}$ is the sample share of bank 1 winners. Hence $f(h)$ is identified everywhere on $[h_{\min}, h_{\max}]$. □

Remark 10 (Support identification). The support bounds $h_{\min}$ and $h_{\max}$ are identified from the range of inferred types among bank 1 winners: $h_{\min} = \tau(r_1^{\min})$ and $h_{\max} = \tau(r_1^{\max})$, where $r_1^{\min}$ and $r_1^{\max}$ are the smallest and largest observed rates among bank 1 winners. This uses the fact that $s(h) > 0$ for all h, so every type has a positive probability of being observed as a bank 1 winner—unlike model 4, where some types might deterministically switch.

5 Overidentification and Testable Restrictions

The model generates testable predictions beyond what is needed for identification.

5.1 Internal Consistency of (6)

Step 3 uses only <u>two</u> types to identify α and λ, but (6) must hold for <u>all</u> types. This generates a continuum of overidentifying restrictions: for any third type h_c,

$$\ln\big(\hat{\alpha}\,m(h_c) - 1\big) \stackrel{?}{=} \hat{\lambda} - \hat{\alpha}\,\Delta(h_c)$$

must hold. Equivalently, a plot of $\ln(\hat{\alpha}\,m(h) - 1)$ against $\Delta(h)$ should lie on a straight line with slope $-\hat{\alpha}$ and intercept $\hat{\lambda}$. Departures from linearity indicate model misspecification (e.g., the logit functional form is wrong, or the equilibrium characterization fails).

5.2 Adverse Selection Pattern

The model predicts (Proposition 3 in the model document):

$$E[D_i \mid W_i = 1] < E[D_i] < E[D_i \mid W_i = 2] \tag{12}$$

Bank 2 winners should have a strictly higher default rate than bank 1 winners. This is directly testable and, importantly, the magnitude of the gap is pinned down by the estimated (α, λ, F).

5.3 Bank 2's First-Order Condition

The entrant's FOC (3) provides an additional restriction. Having identified α, λ, F and $\sigma^*(\cdot)$, the model predicts a specific value of r_2^*. If the directly-observed r_2^* (from bank 2 winners) matches the FOC-implied value, the model is internally consistent. A discrepancy indicates that bank 2 is not best-responding, i.e., the equilibrium assumption fails.

5.4 Default Distribution Among Bank 2 Winners

Among bank 2 winners, the distribution of defaults is a mixture of Bernoulli(h) with mixing density:

$$f_{h|W=2}(h) = \frac{(1 - s(h))\, f(h)}{\Pr(W_i = 2)}$$

which is fully pinned down by the estimates. The implied moments — $E[D|W = 2]$, $\mathrm{Var}(D|W = 2)$, etc. — are testable against the data.

5.5 Switching Rate

The model predicts the aggregate switching probability:

$$\Pr(W_i = 2) = \int_{h_{\min}}^{h_{\max}} \big(1 - s(h)\big)\, dF(h)$$

which can be compared to the observed fraction N_2/N.

6 Likelihood-Based Estimation

6.1 Likelihood Function

For each borrower i, the likelihood contribution depends on which bank wins.

Bank 1 wins ($W_i = 1$): The winning rate is $r_i^{\text{win}} = \sigma^*(h_i)$. Since σ^* is strictly increasing, observing r_i^{win} reveals $h_i = \tau(r_i^{\text{win}})$. The likelihood is:

$$\mathcal{L}_i^{(1)} = \underbrace{f\big(\tau(r_i^{\text{win}})\big) \cdot |\tau'(r_i^{\text{win}})|}_{\text{density of rate}} \cdot \underbrace{s\big(\sigma^*(\tau(r_i^{\text{win}})), r_2^*\big)}_{\text{prob. bank 1 chosen}} \cdot \underbrace{h_i^{D_i}(1 - h_i)^{1-D_i}}_{\text{default probability}} \tag{13}$$

where $h_i = \tau(r_i^{\text{win}})$ throughout, and $|\tau'(r)| = \frac{1}{(\sigma^*)'(\tau(r))}$ is the Jacobian from the change of variables $h \to r$.

Bank 2 wins ($W_i = 2$): The winning rate is $r_i^{\text{win}} = r_2^*$ for all such borrowers. The type h_i is unobserved, so we integrate it out:

$$\mathcal{L}_i^{(2)} = \int_{h_{\min}}^{h_{\max}} \underbrace{\big(1 - s(\sigma^*(h), r_2^*)\big)}_{\text{prob. bank 2 chosen}} \cdot \underbrace{h^{D_i}(1-h)^{1-D_i}}_{\text{default probability}} \cdot f(h)\, dh \tag{14}$$

Full likelihood:

$$\mathcal{L}(\theta) = \prod_{i=1}^{N} \left[\mathcal{L}_i^{(1)}\right]^{\mathbf{1}(W_i=1)} \left[\mathcal{L}_i^{(2)}\right]^{\mathbf{1}(W_i=2)} \tag{15}$$

Remark 11 (Computational structure). For bank 1 winners, the likelihood is in closed form (no integral) because the winning rate reveals the type. For bank 2 winners, a one-dimensional numerical integral over h is required. This asymmetry mirrors model 4's identification, but here both cases involve smooth (logit) probabilities rather than indicator functions, simplifying computation.

6.2 Estimation Procedure

1. **Fix $\theta = (\alpha, \lambda, F$-parameters).** Solve the equilibrium: compute $\sigma^*(h;\theta)$ and $r_2^*(\theta)$ from the system (2)–(3) numerically.[2]
2. **Evaluate the likelihood $\mathcal{L}(\theta)$** using (13)–(14).
3. **Maximize $\ell(\theta) = \sum_i \ln \mathcal{L}_i$** over θ using numerical optimization.
4. **Standard errors** from the inverse Hessian or bootstrap.

7 Discussion

7.1 Role of Default Outcomes

The identification strategy relies critically on observing ex-post default outcomes D_i. Without defaults, the mapping $r_1 \leftrightarrow h$ (Step 2) cannot be constructed nonparametrically: we would observe the distribution of bank 1 rates but not the underlying types generating them. In that case, identification would require stronger parametric assumptions on F and would rely entirely on the equilibrium restrictions to discipline the type distribution.

7.2 Why the Logit Helps Identification (Relative to Model 4)

Compared to model 4 (homogeneous demand, mixed-strategy equilibrium for bank 2), the logit specification offers two advantages for identification:

[2] For each candidate r_2^*, the pointwise FOC (2) defines $\sigma^*(h)$ for each h via a scalar fixed-point problem. Then check whether bank 2's FOC (3) is satisfied; iterate on r_2^* until it is.

1. **Bank 1 winners span the full support.** In model 4, bank 1 wins deterministically when its rate is low enough; in the logit, every type has a positive probability of choosing bank 1 ($s > 0$ for all h). This means the empirical distribution of rates among bank 1 winners covers the full type support, so F is identified everywhere without extrapolation.

2. **No mixed-strategy estimation.** In model 4, bank 2's mixed strategy CDF $G(\cdot)$ must be estimated alongside the structural parameters. In the logit, bank 2 plays a single rate r_2^*—a scalar—which is directly observed. This eliminates an infinite-dimensional nuisance object.

7.3 Exclusion Restriction for α vs. λ

The formal identification (Proposition 6) separates α and λ using variation in h across borrowers, exploiting the differential way each parameter enters the markup equation: α scales the rate gap while λ shifts the level. In practice, this identification may be strengthened by exclusion restrictions:

- Cross-market variation in switching costs (e.g., portability regulation) identifies λ through its effect on market shares, holding α fixed.
- Product-level variation in price sensitivity (e.g., loan size, maturity) can help pin down α separately.

These are not required for point identification (which holds from the model structure alone) but improve finite-sample precision.

7.4 Summary of Identification Logic

Object	**Identified from**	**Step**
r_2^*	Winning rate when $W_i = 2$ (constant)	1
$\sigma^*(h)$, $\tau(r)$	$E[D\|r, W = 1]$ (defaults reveal types)	2
α	Eq. (7): markup curvature across types	3
λ	Eq. (8): markup level given α	3
$F(h)$	Selection-corrected type distribution	4